

Student Result Dashboard - Mini Project Assignment

This assignment focuses on JavaScript DOM Manipulation concepts. Students will dynamically create a Student Result Dashboard using JavaScript only.

1. HTML Structure

```
<body>
  <div id="container"></div>

  <script src="script1.js"></script>
</body>
```

2. CSS (Already Provided)

```
.card{
  padding: 10px;
  margin: 10px;
  border-radius: 8px;
  border: 1px solid #ccc;
}

.topper{
  background-color: #d4edda;
  color: #155724;
  border-left: 6px solid green;
}

.fail{
  background-color: #f8d7da;
  color: #721c24;
  border-left: 6px solid red;
}

.average{
  background-color: #fff3cd;
  color: #856404;
  border-left: 6px solid orange;
}
```

3. Given Student Data

```
var students = [
  { name: "Ali", marks: 85, city: "Karachi" },
  { name: "Ahmed", marks: 72, city: "Lahore" },
  { name: "Sara", marks: 91, city: "Islamabad" },
  { name: "Zain", marks: 60, city: "Multan" }
]
```

4. Project Tasks

1. Select #container using JavaScript.
2. Create a heading:
Student Result Dashboard
and insert it before the container.
3. Create a footer paragraph:
Generated using JavaScript DOM
and insert it after the container.
4. Loop through the students array.
5. For each student:
 - Create a div with class 'card'
 - Create:
 - h2 for name
 - p for marks
 - span for city

6. Apply conditional classes:
 - marks $\geq$ 80 $\rightarrow$ topper
 - marks $<$ 70 $\rightarrow$ fail
 - otherwise $\rightarrow$ average
7. Append h2, p and span inside card.
8. Store all cards inside an array.
9. Render all cards using:
 container.append(...store)
10. Print container in console.

5. Concepts Covered

Concept	Purpose
createElement()	Create HTML elements dynamically
append()	Insert elements into DOM
before() / after()	Insert elements before or after another element
classList.add()	Add CSS classes dynamically
Arrays + Loop	Render data dynamically
Conditional Logic	Apply different classes based on marks
Spread Operator	Append all cards in one DOM update

Focus mainly on JavaScript DOM Manipulation. CSS is already provided to save time and keep focus on JavaScript logic.